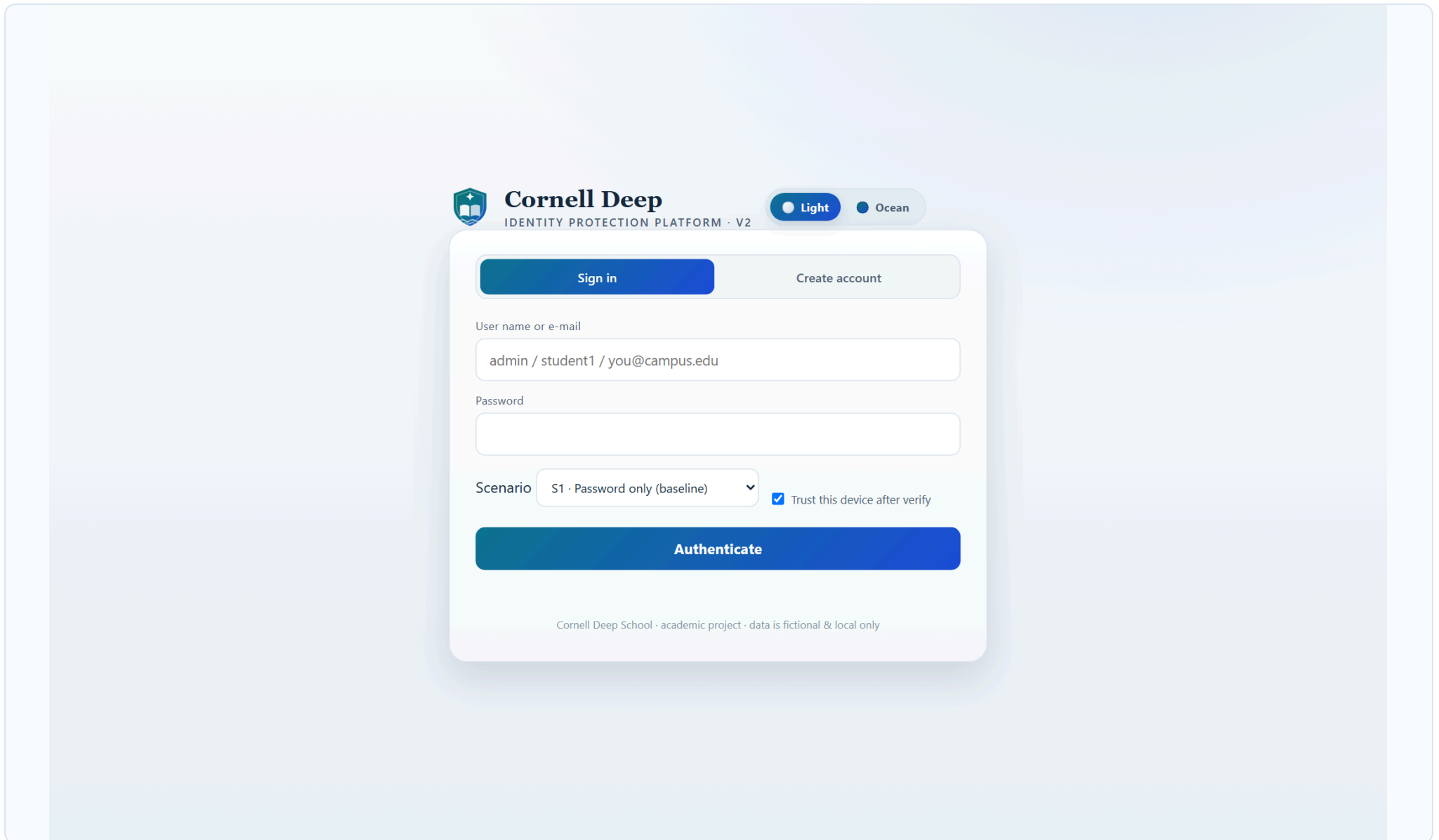


1 / 15 Login screen — default Light theme, with the toggle to Ocean



Login screen — default Light theme, with the toggle to Ocean — 01-login-light.png

2 / 15 Login screen in the Ocean theme (green/blue) — same content, different visual identity

The screenshot displays the login interface for the Cornell Deep Identity Protection Platform (V2) in the Ocean theme. The background is a light blue gradient. At the top left is the Cornell Deep logo, featuring a shield with a book and a star, followed by the text "Cornell Deep" and "IDENTITY PROTECTION PLATFORM · V2". To the right of the logo are two theme toggle buttons: "Light" (inactive) and "Ocean" (active, highlighted in dark blue). Below the logo, there are two buttons: "Sign in" (dark blue) and "Create account" (light gray). The main form area contains a "User name or e-mail" field with the text "admin / student1 / you@campus.edu", a "Password" field, and a "Scenario" dropdown menu set to "S1 · Password only (baseline)". To the right of the dropdown is a checked checkbox labeled "Trust this device after verify". Below these fields is a large dark blue "Authenticate" button. At the bottom of the form, a small line of text reads "Cornell Deep School · academic project · data is fictional & local only".

Login screen in the Ocean theme (green/blue) — same content, different visual identity — 02-login-ocean.png

3 / 15 Seed accounts for testing with a live Demo OTP code for each account

Cornell Deep
IDENTITY PROTECTION PLATFORM · V2

Light Ocean

Sign in Create account

User name or e-mail

admin / student1 / you@campus.edu

Password

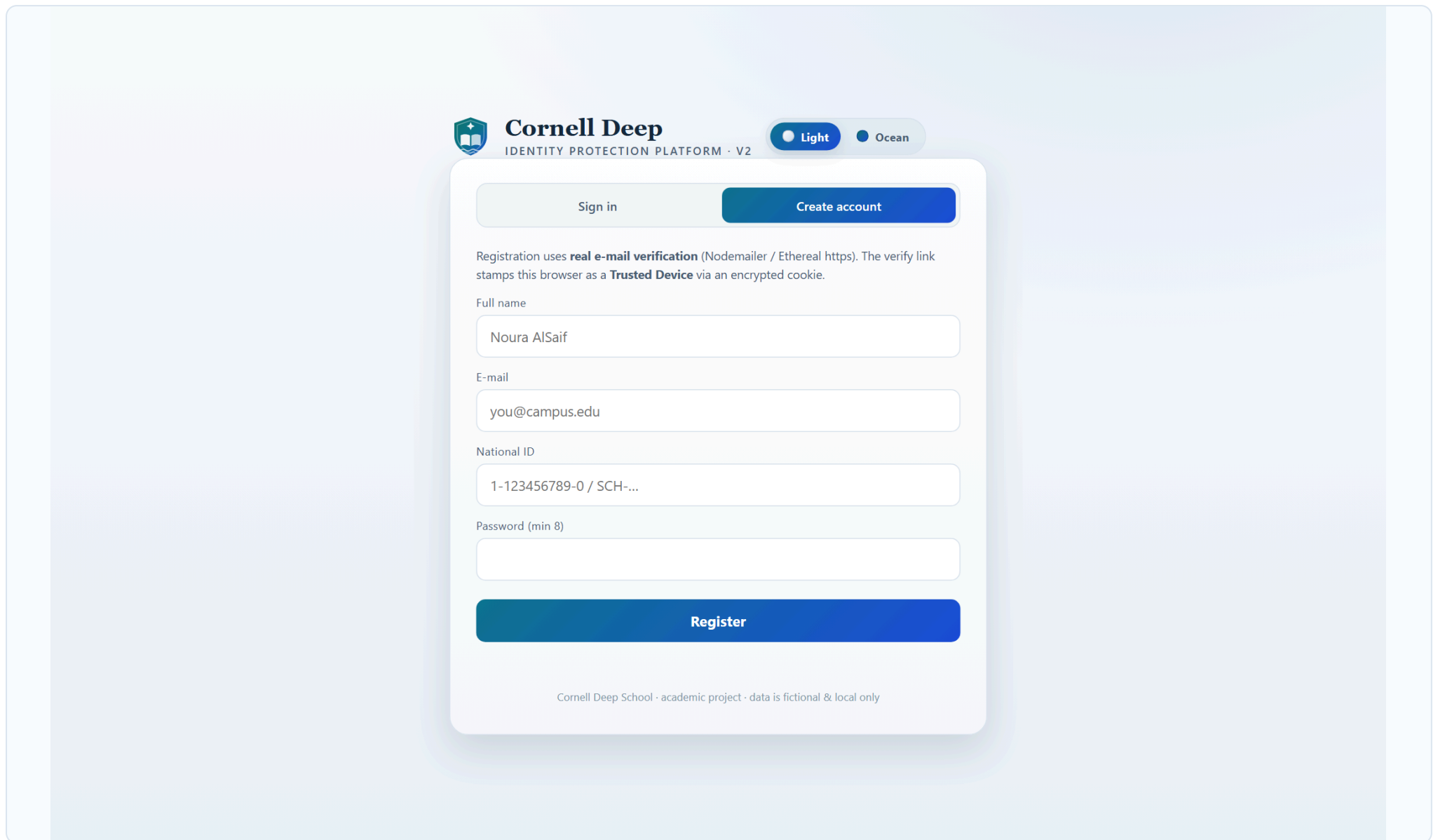
Scenario S1 · Password only (baseline) ☒ Trust this device after verify

Authenticate

Cornell Deep School · academic project · data is fictional & local only

Seed accounts for testing with a live Demo OTP code for each account — 03-login-seed-accounts.png

4 / 15 Creating a new account then verifying e-mail — start of the registration journey



The screenshot shows a web interface for the Cornell Deep Identity Protection Platform. At the top, there is a logo for Cornell Deep and a toggle switch for 'Light' and 'Ocean' themes. Below the logo, there are two buttons: 'Sign in' and 'Create account'. A paragraph of text explains the registration process, mentioning 'real e-mail verification' and 'Trusted Device'. The form consists of four input fields: 'Full name' (containing 'Noura AlSaif'), 'E-mail' (containing 'you@campus.edu'), 'National ID' (containing '1-123456789-0 / SCH-...'), and 'Password (min 8)'. A large blue 'Register' button is positioned below the password field. At the bottom of the form, a small disclaimer states: 'Cornell Deep School · academic project · data is fictional & local only'.

Cornell Deep
IDENTITY PROTECTION PLATFORM · V2

Light Ocean

Sign in Create account

Registration uses **real e-mail verification** (Nodemailer / Ethereum https). The verify link stamps this browser as a **Trusted Device** via an encrypted cookie.

Full name

Noura AlSaif

E-mail

you@campus.edu

National ID

1-123456789-0 / SCH-...

Password (min 8)

Register

Cornell Deep School · academic project · data is fictional & local only

Creating a new account then verifying e-mail — start of the registration journey — 04-register.png

 **Cornell Deep**
ADMINISTRATOR

Portal

Security Center

SOC Console

Forensics

Benchmark

 **Cornell Deep**
SECURITY CENTER

Overview

Audit Log

Sessions

Trusted Devices

Threat Factors

Blacklist

Attack Simulation

Defense Orchestrator ADMINISTRATOR

1

SUCCESS

0

FAILED

0

INVALID OTP

0

BLOCKED IPS

0

STEP-UP

0

ALERTS

0

SSE LIVE

Threat overview

LOCKOUT POLICY
2 consecutive failures → immediate block

RISK ENGINE
LOW ≤30 · MED 31–69 · HIGH ≥70

CROSS-DEVICE APPROVAL
WebSocket/SSE push · 45 s TTL · OTP fallback

THREAT FEED
1 recent events

updated 09:39:52 PM

Realtime security feed

LIVE

TIME	OUTCOME	USER	ROLE	RISK	IP · NETWORK	REASONS / MFA
09:39:52 PM	SUCCESS	admin	ADMINISTRATOR	55	127.0.0.1 · Localhost Loopback	NEW_IP, UNRECOGNIZED_DEVICE

Attack simulation timeline

idle

Run an attack from the **Attack Simulation** tab.

6 / 15 Live authentication log — every attempt with its risk score, factors and reason

**Cornell Deep**
ADMINISTRATOR

PortalSecurity CenterSOC ConsoleForensicsBenchmark

**Cornell Deep**
SECURITY CENTER

Defense OrchestratorADMINISTRATOR

Overview

Audit Log

Sessions

Trusted Devices

Threat Factors

Blacklist

Attack Simulation

1
SUCCESS

0
FAILED

0
INVALID OTP

0
BLOCKED IPS

0
STEP-UP

0
ALERTS

1
SSE LIVE

Audit log

outcome: all▼role: all▼user filterFilterClear threat factors

TIME	OUTCOME	ACTOR	RISK	DETAIL
Sep 28, 09:39:52 PM	SUCCESS	admin (ADMINISTRATOR)	55	NEW_IP, UNRECOGNIZED_DEVICE

Live authentication log — every attempt with its risk score, factors and reason — 06-soc-audit-log.png

7 / 15 Active sessions — device, network, and last activity time

**Cornell Deep**
ADMINISTRATOR

PortalSecurity CenterSOC ConsoleForensicsBenchmark

**Cornell Deep**
SECURITY CENTER

OverviewAudit LogSessionsTrusted DevicesThreat FactorsBlacklistAttack Simulation

Defense OrchestratorADMINISTRATOR

1
SUCCESS

0
FAILED

0
INVALID OTP

0
BLOCKED IPS

0
STEP-UP

0
ALERTS

1
SSE LIVE

Active sessions

terminate any session remotely

USER	IP	NETWORK	MFA	STARTED	
admin (ADMINISTRATOR)	127.0.0.1		—	Sep 28, 09:39:52 PM	<button>Terminate</button>

Active sessions — device, network, and last activity time — 07-soc-sessions.png

8 / 15 Known devices + trust and block lists (whitelist / blacklist)

**Cornell Deep**
ADMINISTRATOR

PortalSecurity CenterSOC ConsoleForensicsBenchmark

**Cornell Deep**
SECURITY CENTER

OverviewAudit LogSessions**Trusted Devices**Threat FactorsBlacklistAttack Simulation

Defense OrchestratorADMINISTRATOR

1
SUCCESS

0
FAILED

0
INVALID OTP

0
BLOCKED IPS

0
STEP-UP

0
ALERTS

1
SSE LIVE

Trusted devices

encrypted trust cookie · 1-year validity

USER	DEVICE	BROWSER	FINGERPRINT	LAST SEEN
user #usr-3001	Edge · Campus PC	Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) HeadlessChrome/154.0.0.0 Safari/537.36 Edg/154.0.0.0	fp-769eb667-18c	Sep 28, 09:39:52 PM

Known devices + trust and block lists (whitelist / blacklist) — 08-soc-devices.png

9 / 15 Attack simulation (SME) — types A/B/C and their instant impact on SOC

**Cornell Deep**
ADMINISTRATOR

PortalSecurity CenterSOC ConsoleForensicsBenchmark

**Cornell Deep**
SECURITY CENTER

OverviewAudit LogSessionsTrusted DevicesThreat FactorsBlacklistAttack Simulation

Defense OrchestratorADMINISTRATOR

1
SUCCESS

0
FAILED

0
INVALID OTP

0
BLOCKED IPS

0
STEP-UP

0
ALERTS

1
SSE LIVE

Attack simulation

runs against the REAL auth risk pipeline

Type A · Brute-Force Spray

Launch attack

Live attack telemetry

Run an attack to watch the SOC respond in real time.

Past runs

ID	TITLE	STATUS	EVENTS
----	-------	--------	--------

Attack simulation (SME) — types A/B/C and their instant impact on SOC — 09-soc-attack-simulation.png



Cornell Deep
ADMINISTRATOR

[Portal](#) [Security Center](#) [SOC Console](#) [Forensics](#) [Benchmark](#)

Light

Ocean

Alex Morgan [Log out](#)

Authentication Mode Comparison

SRS 1.2 — measured from the real audit trail

MODE	ATTEMPTS	AVG LOGIN TIME	SUCCESS	FAILED PASSWORD	2ND-FACTOR FAILURES	BLOCKED	STEP-UPS
Password-only baseline (S1)	0	—	0	0	0	0	0
Password + Mobile OTP MFA (S2)	0	—	0	0	0	0	0
Password + OTP + E-mail step-up + RBAC (S3)	1	25 ms	1	0	0	0	0

 Measured from real audit trail (elapsed_ms) — run each scenario ≥3 times for a stable average.

Security / Usability profile (SRS Table 1)

MODE	SECURITY STRENGTH	USABILITY	RISK REDUCED WHEN...
Password-only	Low–Medium	Very easy	adds no protection if the password is stolen
Password + OTP	Medium–High	Easy–moderate	a stolen password alone is insufficient
Password + OTP + E-mail step-up	Higher	Moderate (depends on e-mail access)	an additional barrier if the registered e-mail stays uncompromised

Scenario 1/2/3 comparison from a real log (attempts, success, time) — 10-benchmark-scenarios.png

11 / 15 Timeline reconstruction and opening a case with automatic evidence capture

 **Cornell Deep**
ADMINISTRATOR

PortalSecurity CenterSOC ConsoleForensicsBenchmark

LightOcean

Alex MorganLog out

Digital Forensic Investigations

timeline reconstruction from the persisted database

CASES
0

AUTH EVENTS
1

ALERTS
0

SUCCESSFUL LOGINS
1

ATTACK RUNS
0

TRUSTED DEVICES
1

Reconstruct timeline

correlated chain across audit · sessions · alerts · approvals · simulations

outcome: all

SearchClear

Search any user, IP or device to reconstruct the event chain.

Open a case

wrap a scope into an investigation · auto-captures matching evidence

CRITICAL

scope: username

scope: IP

scope: device

Create case

Cases

REF	TITLE	SEVERITY	STATUS	EVENTS	CREATED
No cases yet — open one above.					

Timeline reconstruction and opening a case with automatic evidence capture — 11-forensics-timeline.png

12 / 15 Student personal Security Center + recovery / MFA reset path

**Cornell Deep**
STUDENT

Portal **Security Center**

Light

Ocean

Sarah Chen Log out

My Security Center **STUDENT**

1

TRUSTED DEVICES

1

ACTIVE SESSIONS

1

ACTIVITY

E-MAIL VERIFIED
✔ Verified

LAST LOGIN RISK POSTURE
recovery enabled · self-service

NATIONAL ID
SCH-2019001345

My trusted devices

DEVICE	BROWSER	FINGERPRINT	LAST SEEN
Edge · Campus PC	Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) HeadlessChrome/154.0.0.0 Safari/537.36 Edg/154.0.0.0	fp-769eb667-18c	Sep 28, 09:40:13 PM

Revoke

My sessions

IP	NETWORK	MFA	STARTED
127.0.0.1		—	Sep 28, 09:40:13 PM

this session

Recent activity

TIME	OUTCOME	RISK	REASONS
Sep 28, 09:40:13 PM	SUCCESS	55	NEW_IP, UNRECOGNIZED_DEVICE

Account recovery · MFA reset

self-service via registered identity

username or e-mail

Request recovery code

username to validate

recovery code

Validate recovery code

Student personal Security Center + recovery / MFA reset path — 12-student-security-center.png

**Cornell Deep**
STUDENT

PortalSecurity Center

LightOcean

Sarah ChenLog out

School portal

fictional modules · role: STUDENT · 3 courses · dummy data

Overview

Courses

Grades & Exams

Assignments

Attendance

COURSES

3

GRADES

3

ASSIGNMENTS

4

ATTENDANCE

4

Your academic overview — grades, assignments and attendance for your enrolled courses.

RBAC enforced server-side: every module request is scoped to your STUDENT session.

School portal — student view: courses, grades and attendance (RBAC) — 13-portal1-student.png

14 / 15 School portal — teacher view: courses, rosters and module links (RBAC)

Cornell Deep
TEACHER

Portal Security Center

Light Ocean Dr. Elena Roberts Log out

School portal

fictional modules · role: TEACHER · 2 courses · dummy data

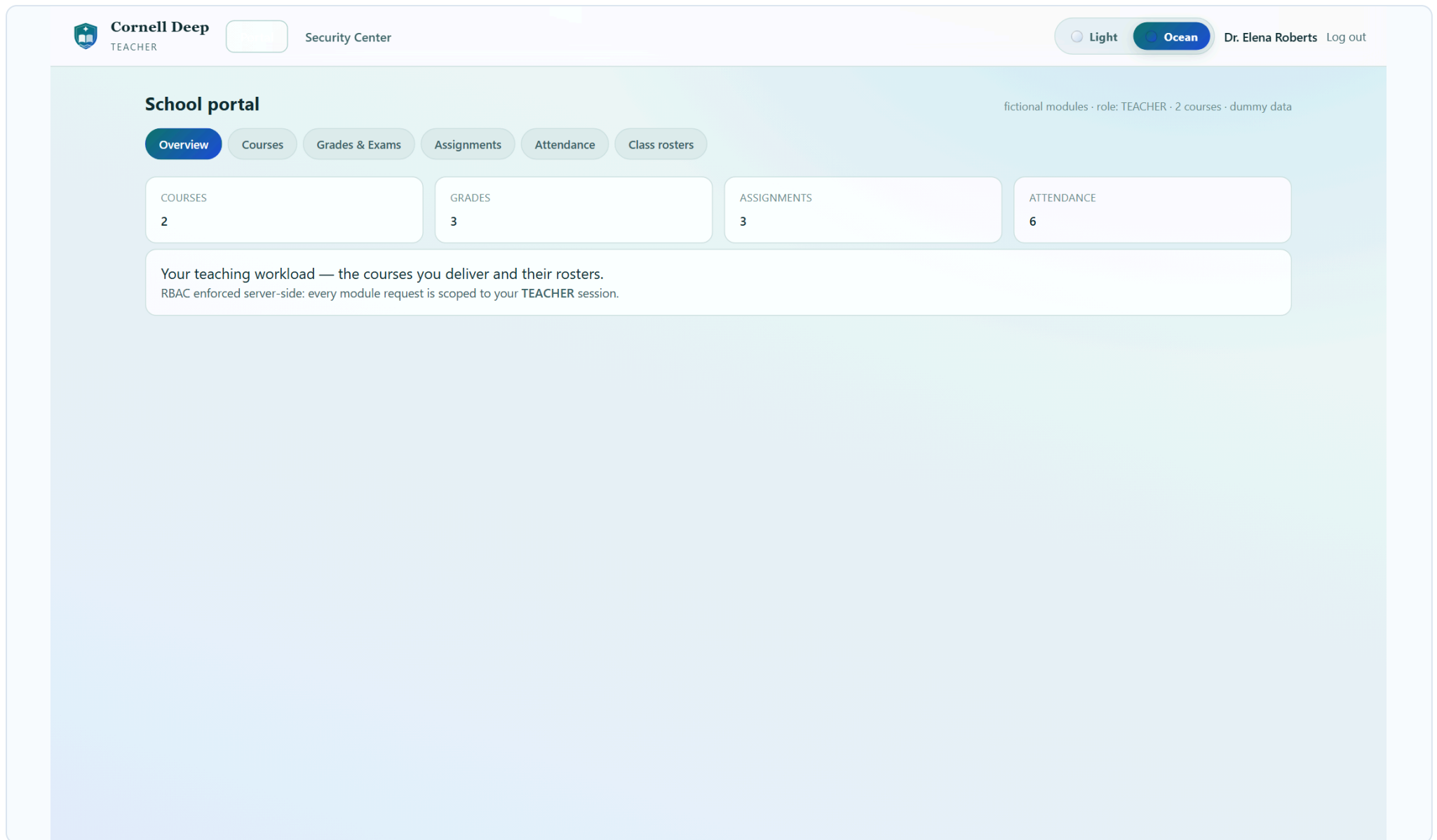
Overview Courses Grades & Exams Assignments Attendance Class rosters

COURSES 2	GRADES 3	ASSIGNMENTS 3	ATTENDANCE 6
--------------	-------------	------------------	-----------------

Your teaching workload — the courses you deliver and their rosters.
RBAC enforced server-side: every module request is scoped to your **TEACHER** session.

School portal — teacher view: courses, rosters and module links (RBAC) — 14-portal1-teacher.png

15 / 15 The app after login in the Ocean theme — theme persists between sessions



The app after login in the Ocean theme — theme persists between sessions — 15-app-ocean-theme.png